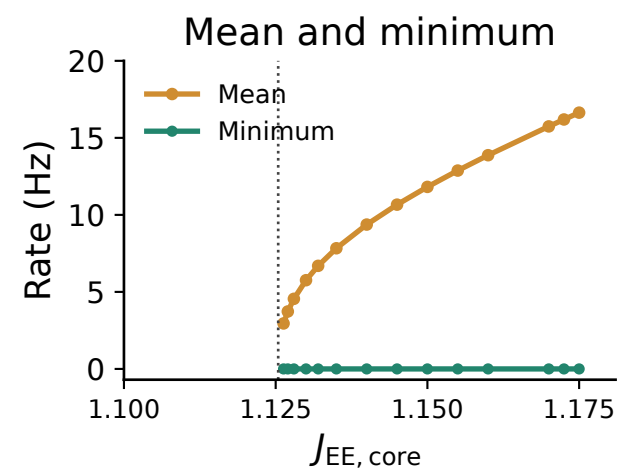
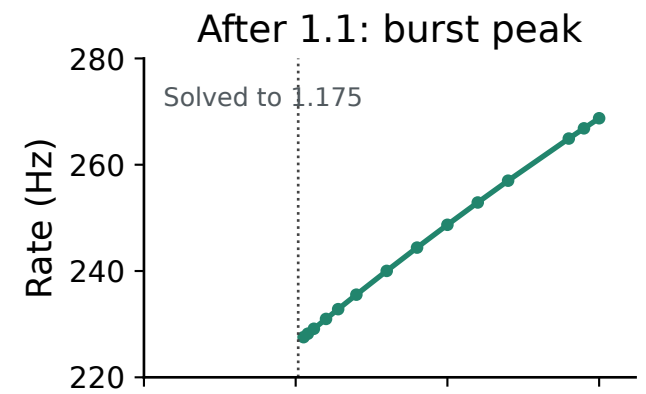
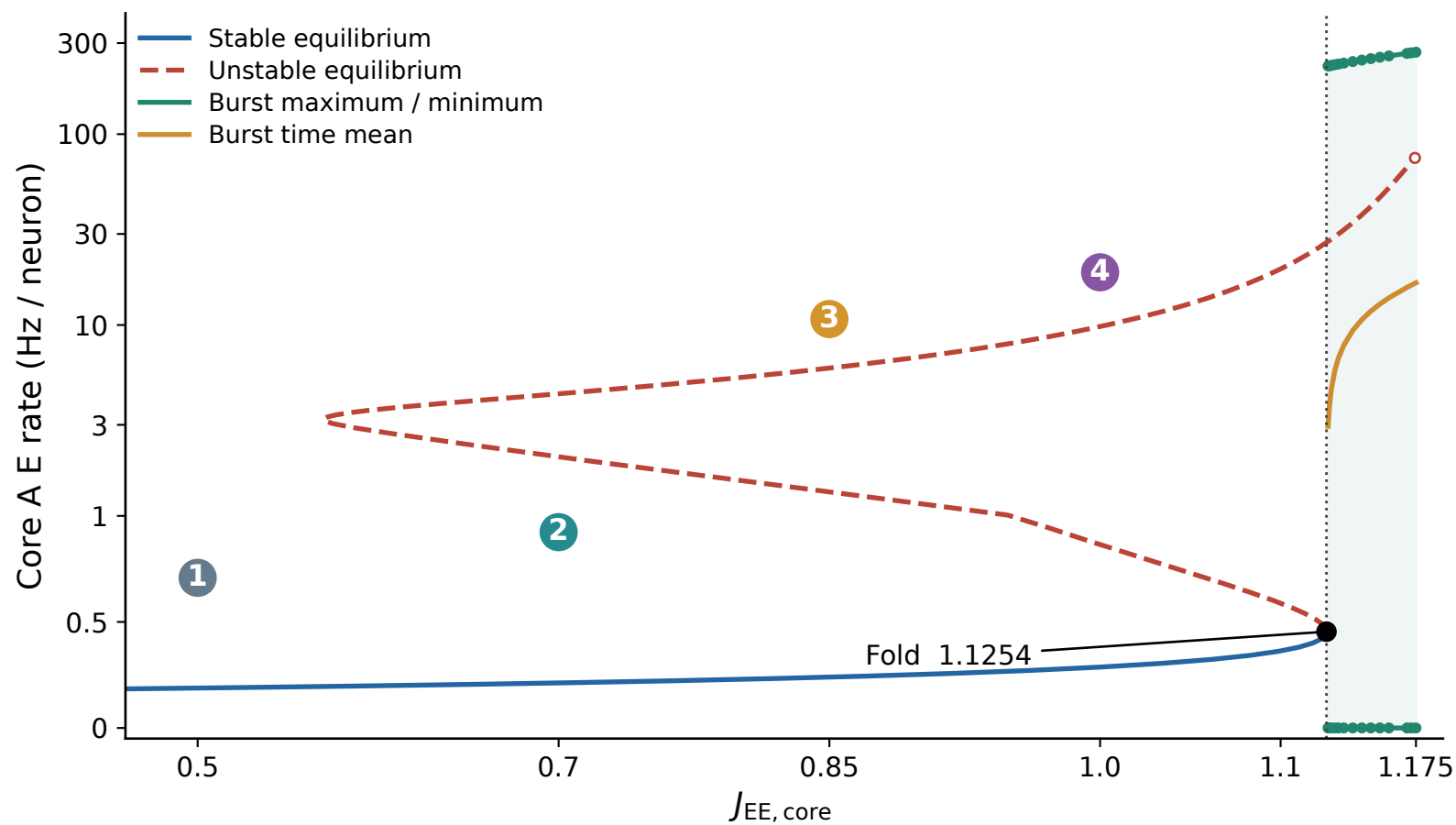


Core burst onset and four native SNN examples

Deterministic bifurcation branches; numbered circles are native SNN time means.

Equilibria and stable burst orbits



1-4: the same Core A, threshold field and noise seed; all four precede the closure fold.

1 Resting

$J_{EE,core} = 0.5$ | no bursts

2 Irregular bursts

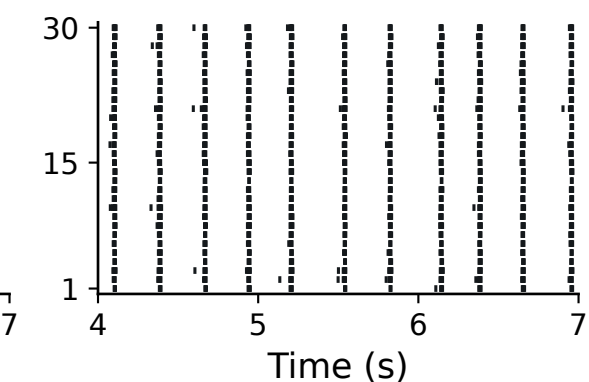
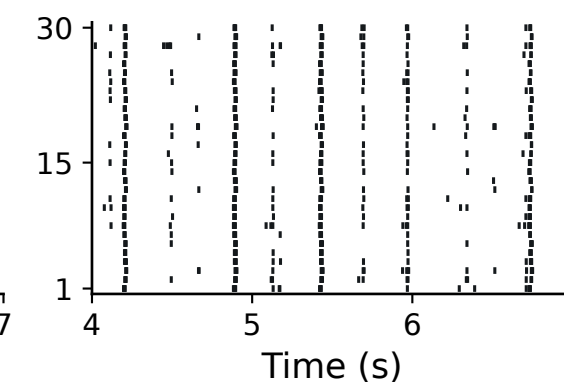
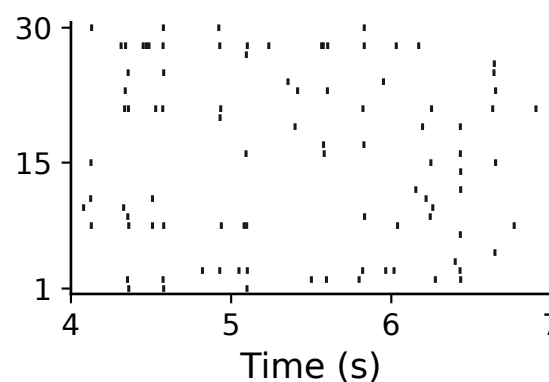
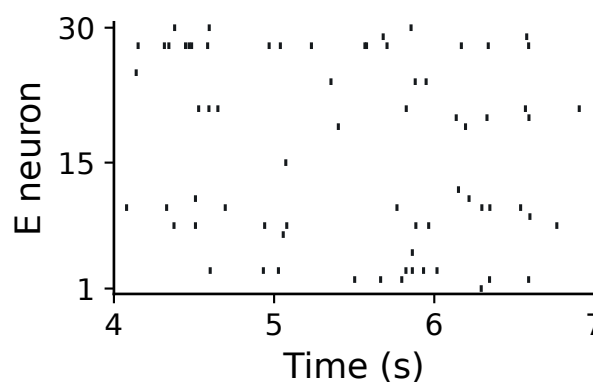
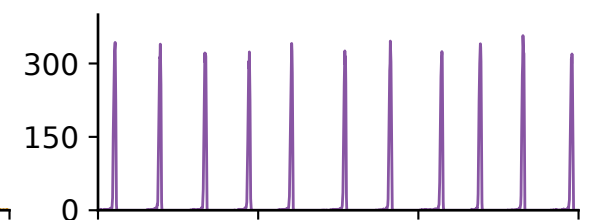
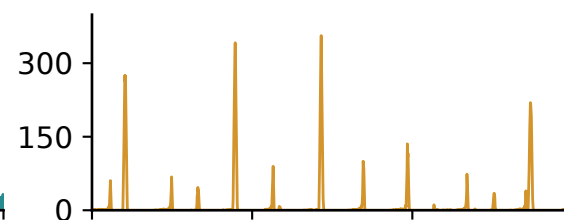
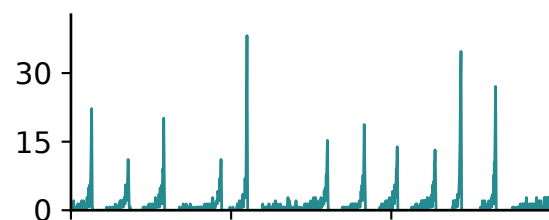
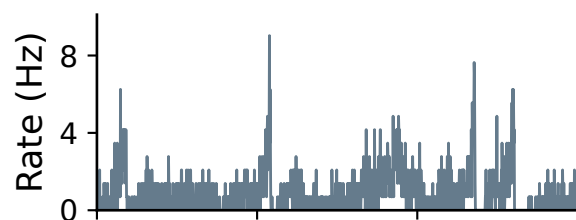
$J_{EE,core} = 0.7$ | IEI CV 0.70

3 Intermediate

$J_{EE,core} = 0.85$ | IEI CV 0.35

4 Regular bursts

$J_{EE,core} = 1$ | IEI CV 0.13



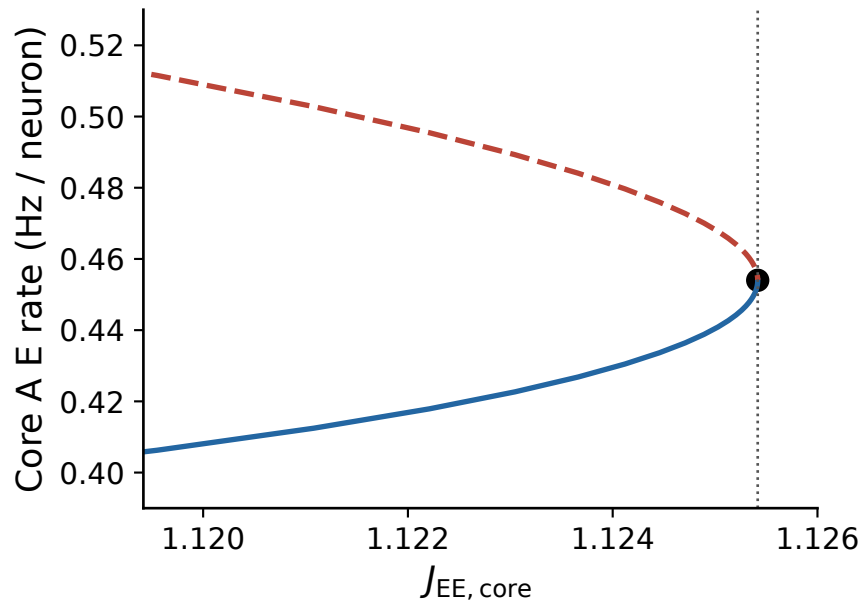
Waveforms: all 720 Core A E cells; linear rate axes vary by panel. Raster: the same 30 cells, occupied 2 ms bins.

IEI CV uses 2-20 s; displayed window is 4-7 s. Native labels do not identify deterministic attractors.

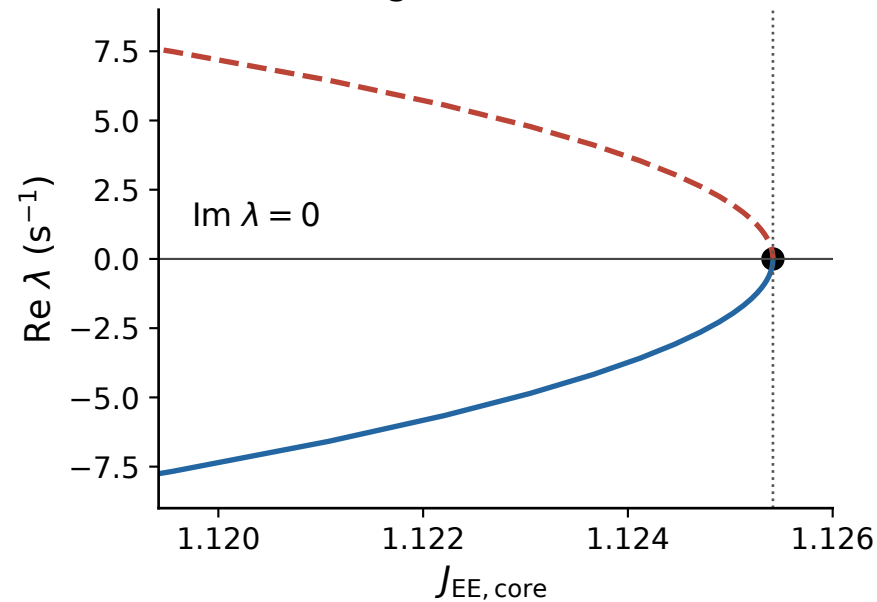
The low-rate fold and its critical mode

$J_{EE, \text{core}} = 1.12541641$ | Core A E = 0.454005 Hz

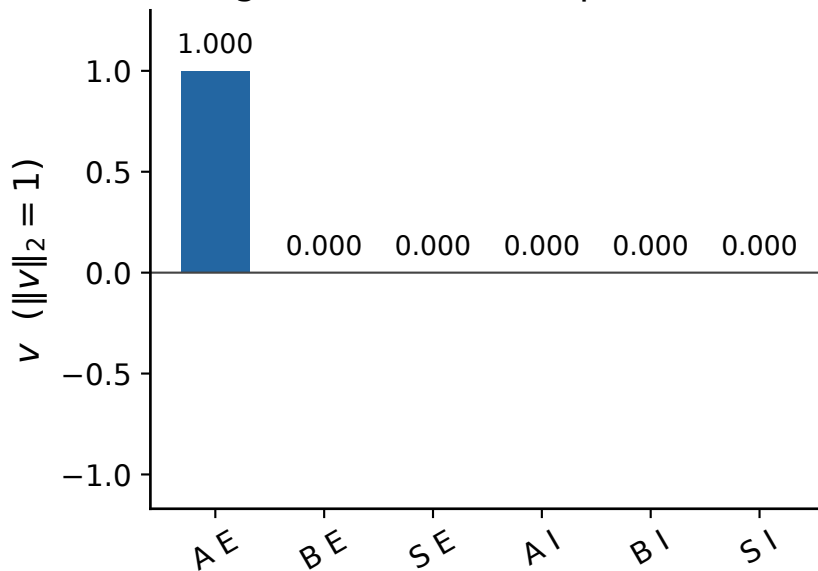
Stable and saddle branches meet



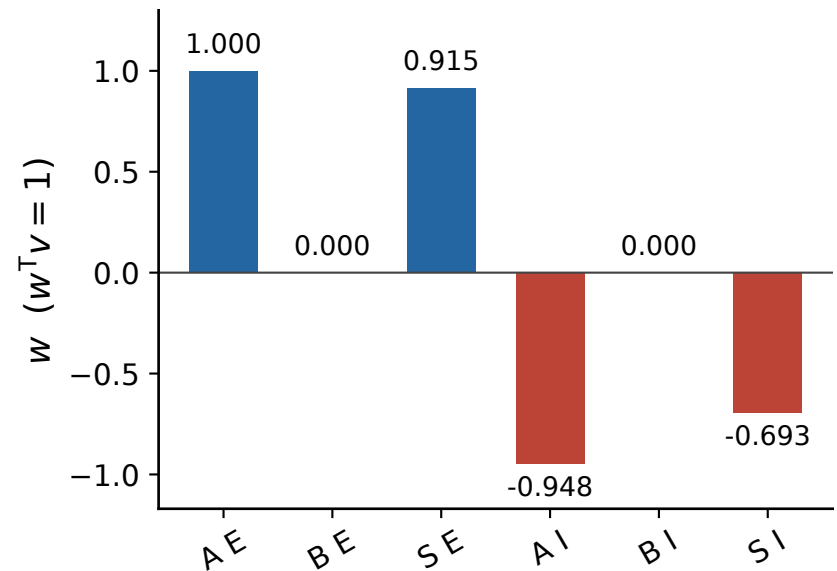
One real eigenvalue reaches zero



Right mode: rate displacement



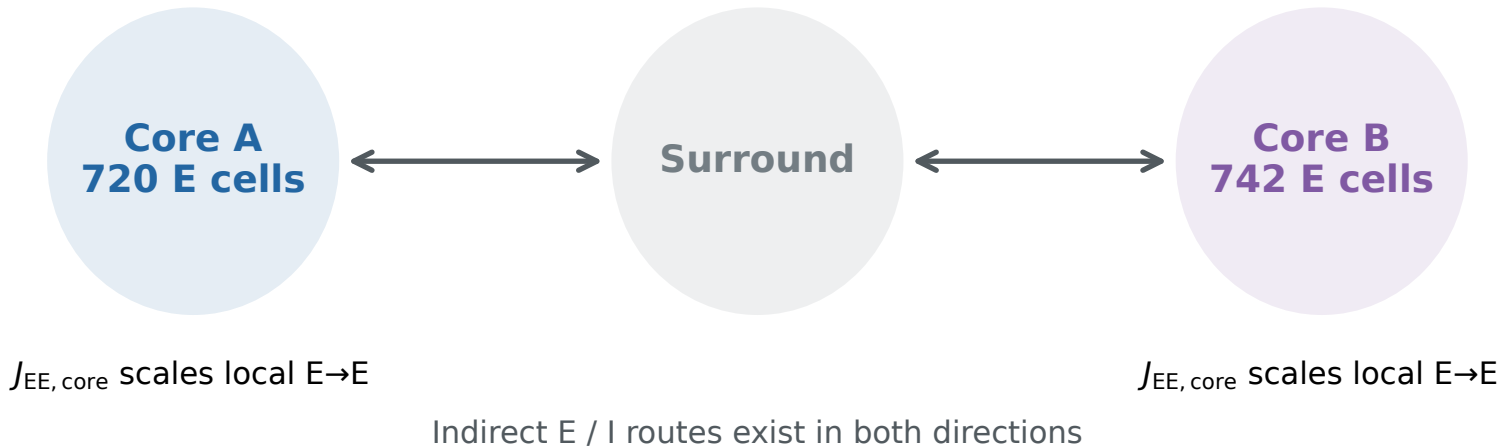
Left characteristic vector



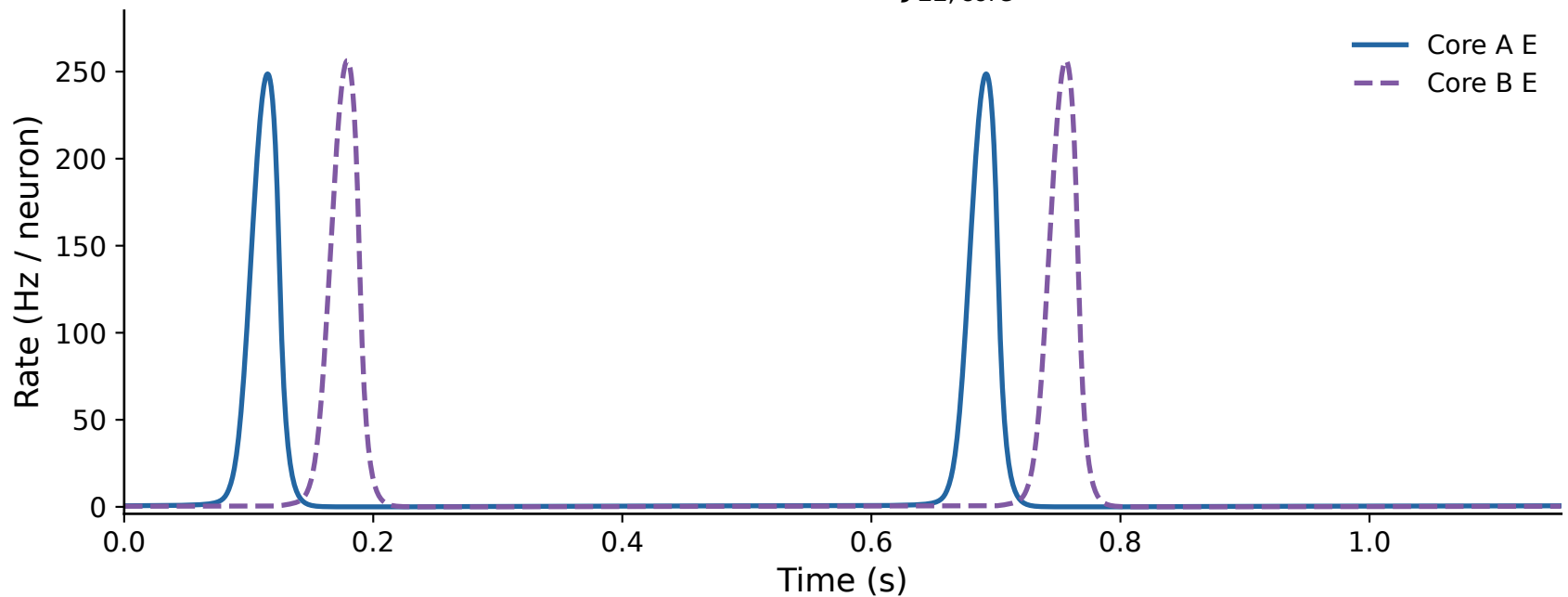
Six-population characteristic vectors; S = surround. Full 402-coordinate vectors are retained in v2.

Core A and B: two spatial populations

No direct A↔B edges in this realized graph



Solved burst orbit at $J_{EE, \text{core}} = 1.15$



Full-network period: 577.009 ms. B peak follows A peak by 63.96 ms on this orbit.

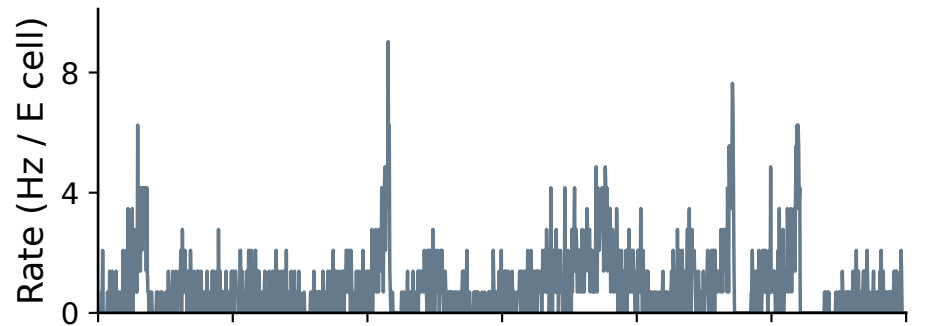
The fold mode begins in A; the later burst recruits both cores. A/B are spatial identities, not state labels.

Diagram summarizes existing graph routes; peak order alone does not establish causal direction.

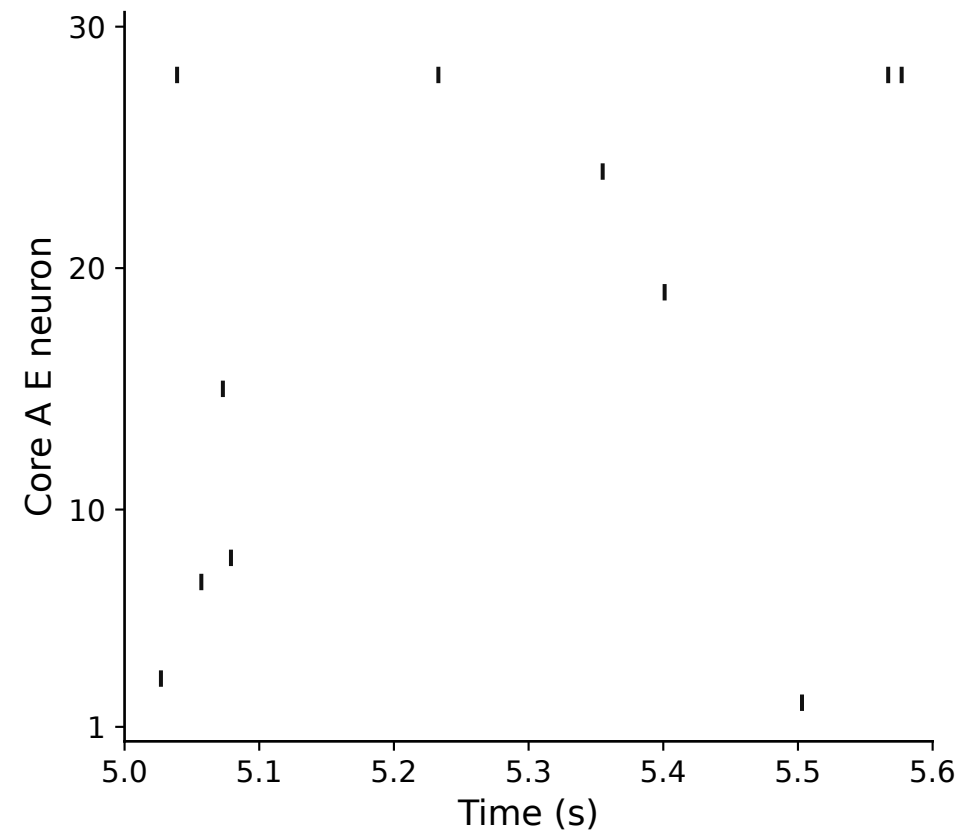
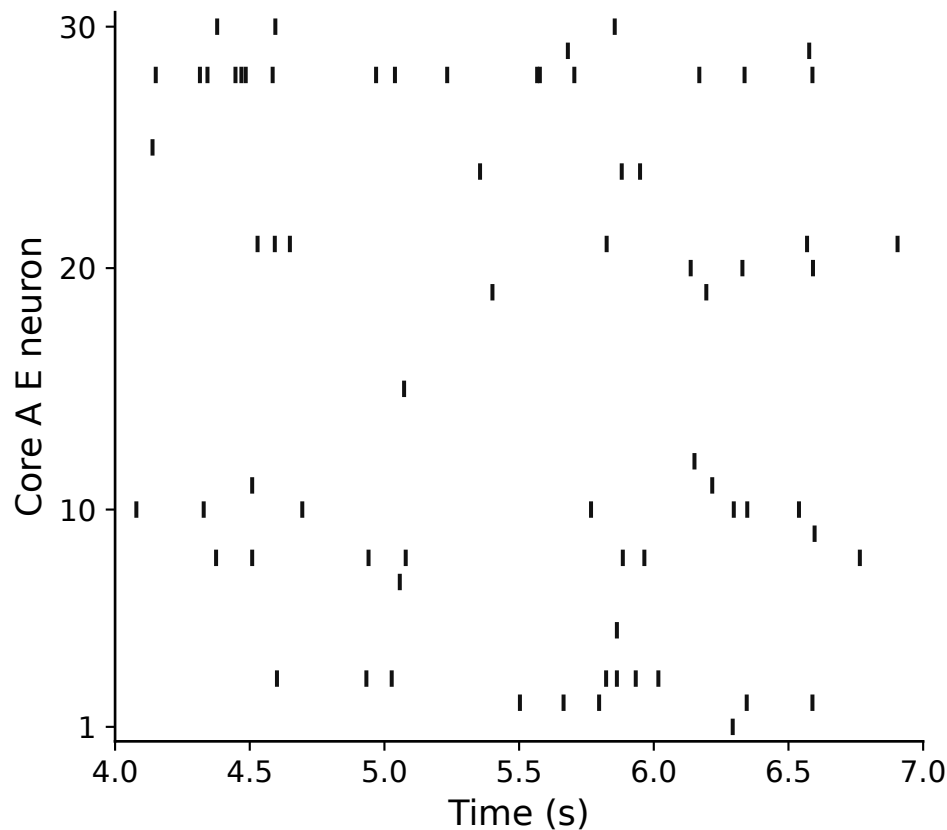
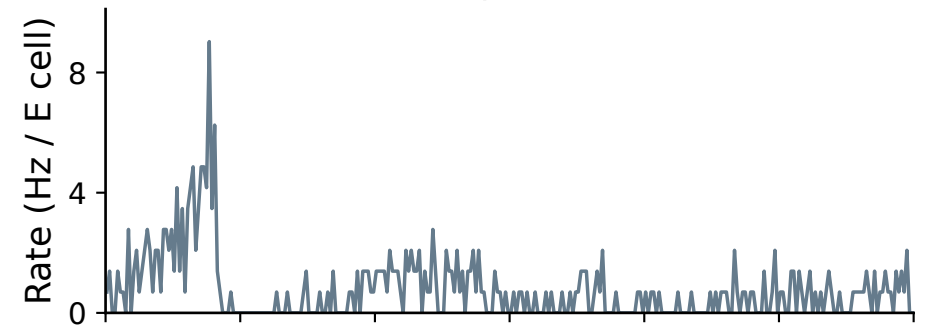
1 Resting / low activity

$J_{EE, core} = 0.5$ | No detected bursts | mean = 0.708 Hz

Same 3-second window



Same 600-ms close-up

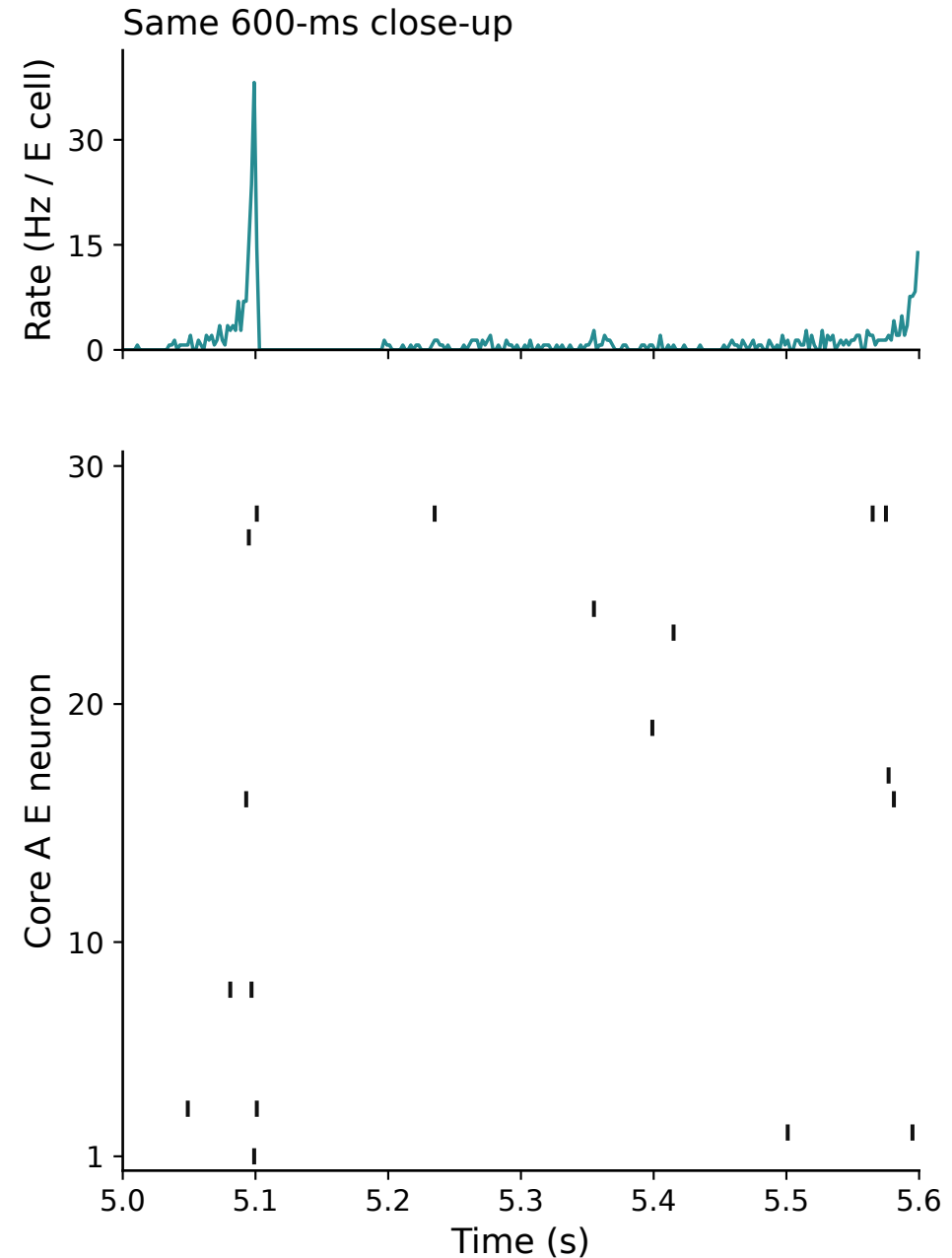
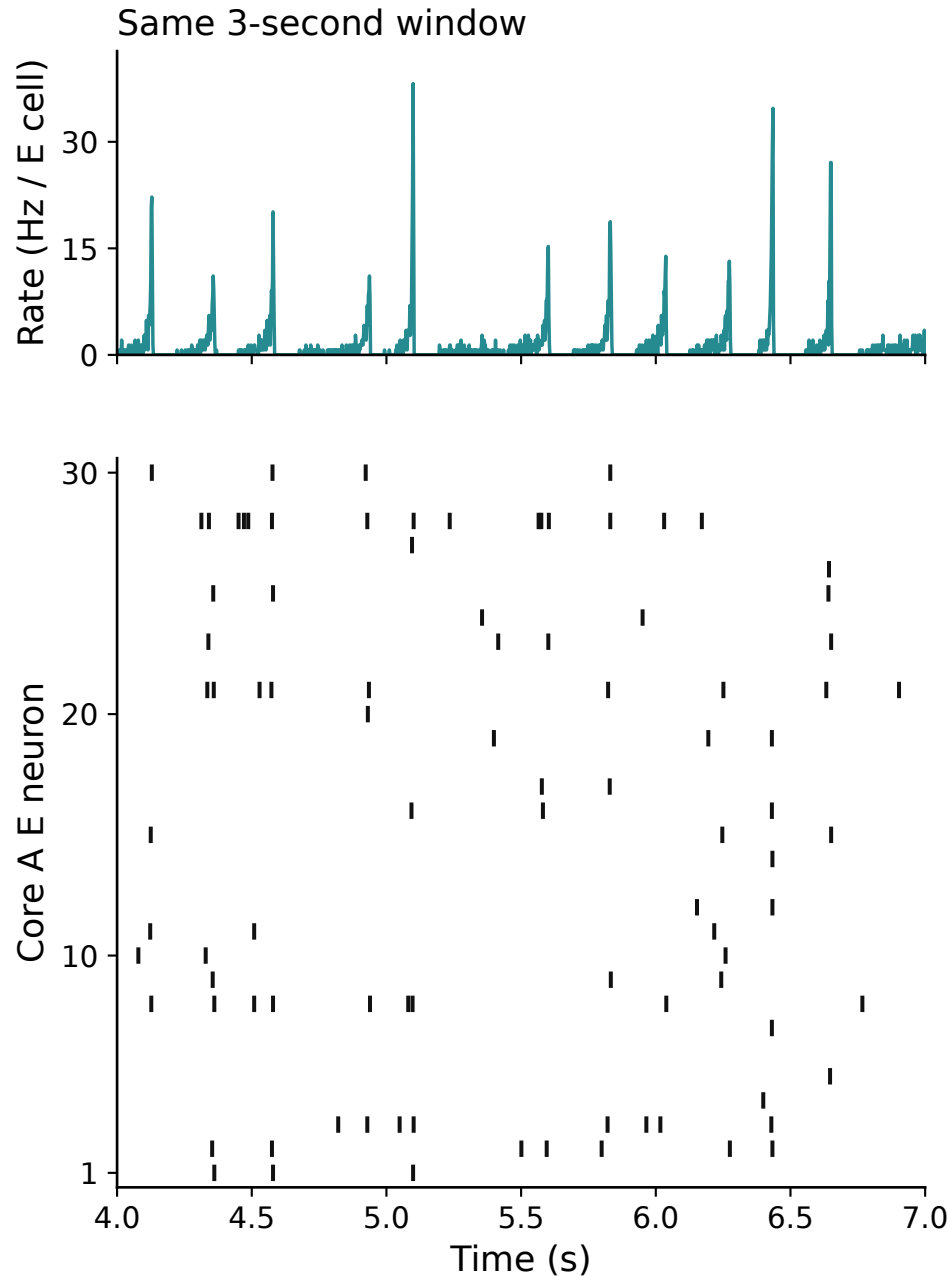


Native SNN. Rate: all 720 Core A E cells. Raster: occupied 2 ms bins of the same 30 recorded cells.

Metrics: 2-20 s; fixed topology 2511, threshold amplitude 1 and noise seed 848101.

2 Irregular bursts

$J_{EE, core} = 0.7$ | $IEI\ CV = 0.699$; 29 bursts | mean = 0.923 Hz

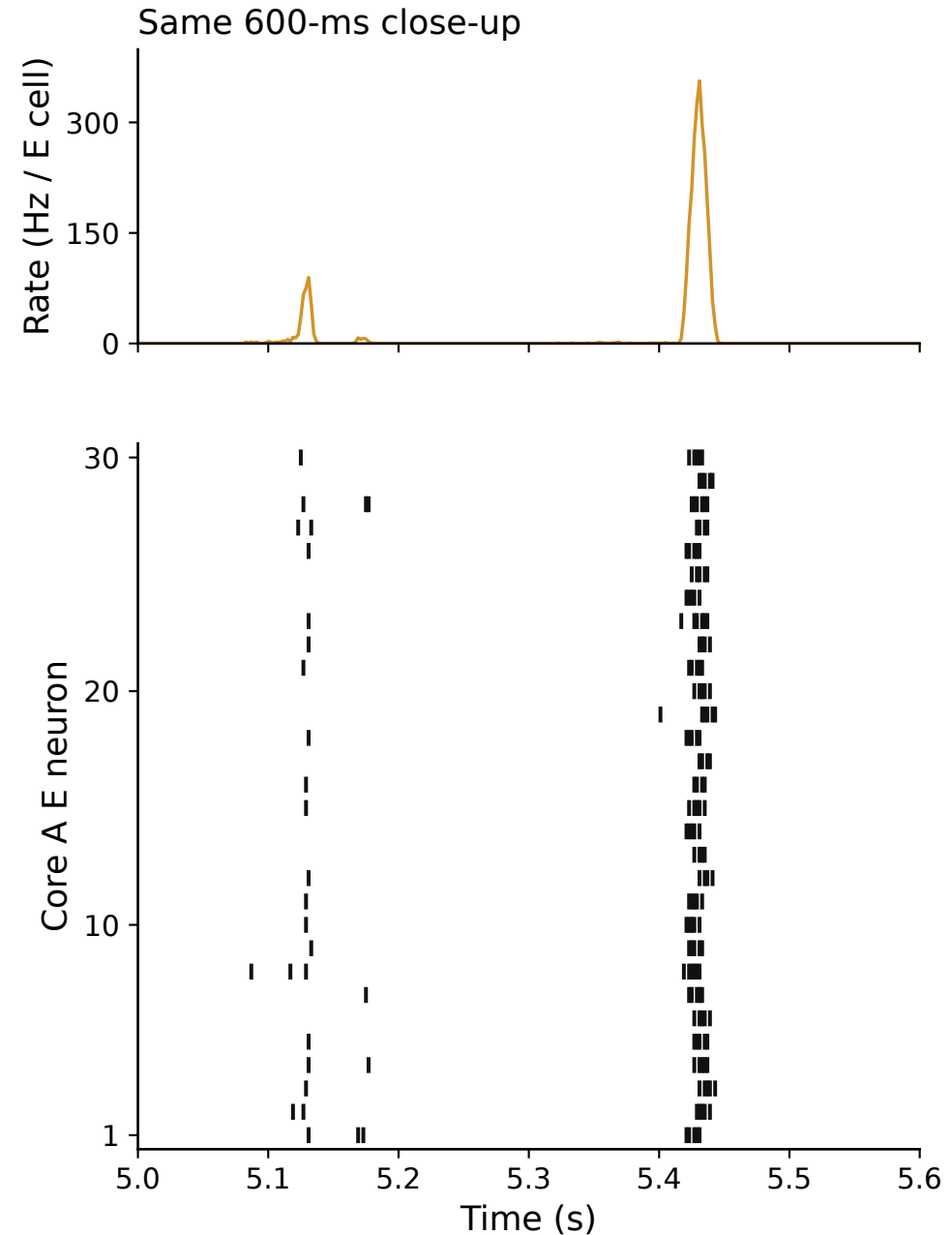
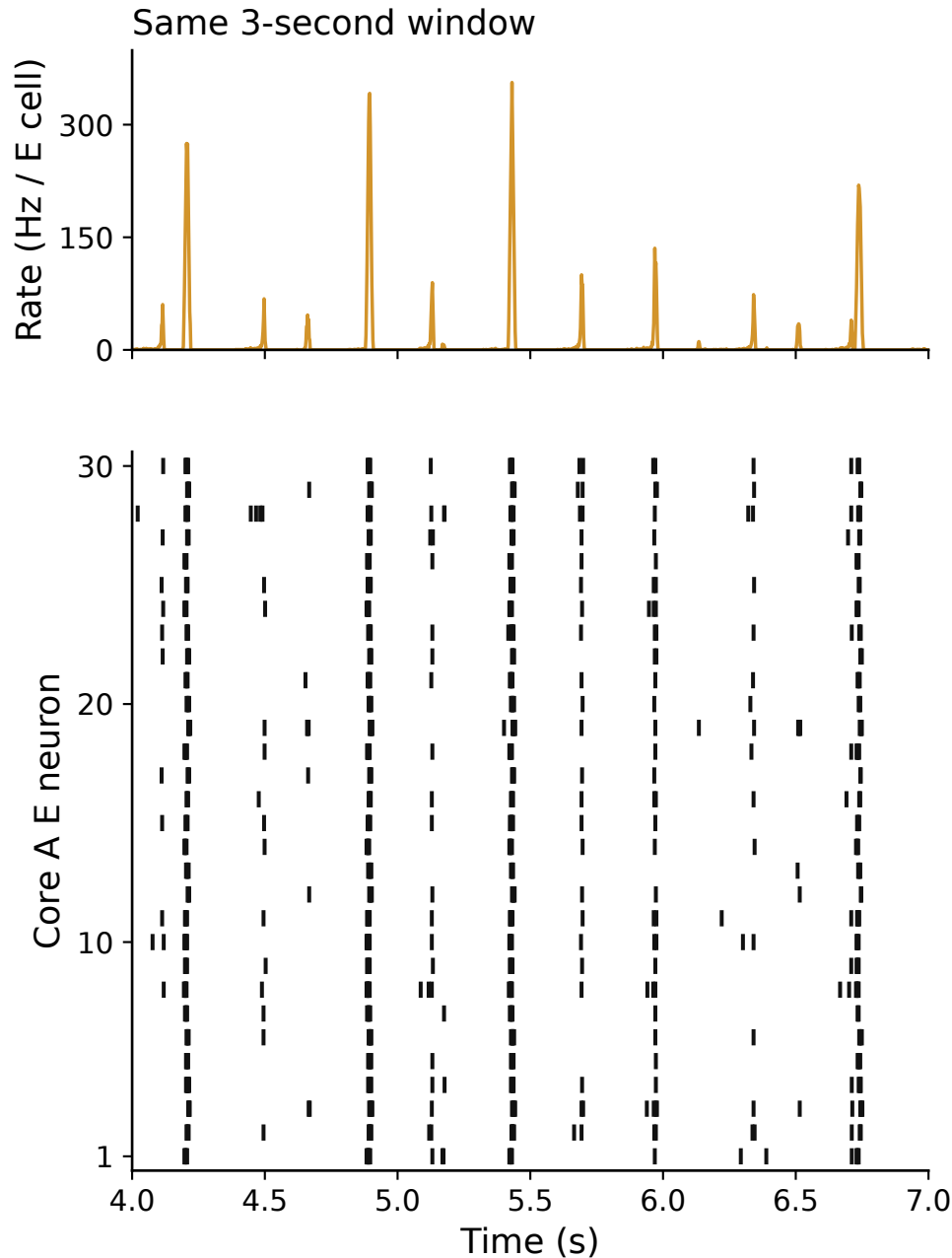


Native SNN. Rate: all 720 Core A E cells. Raster: occupied 2 ms bins of the same 30 recorded cells.

Metrics: 2-20 s; fixed topology 2511, threshold amplitude 1 and noise seed 848101.

3 Intermediate

$J_{EE, core} = 0.85$ | IEI CV = 0.345; 63 bursts | mean = 10.747 Hz

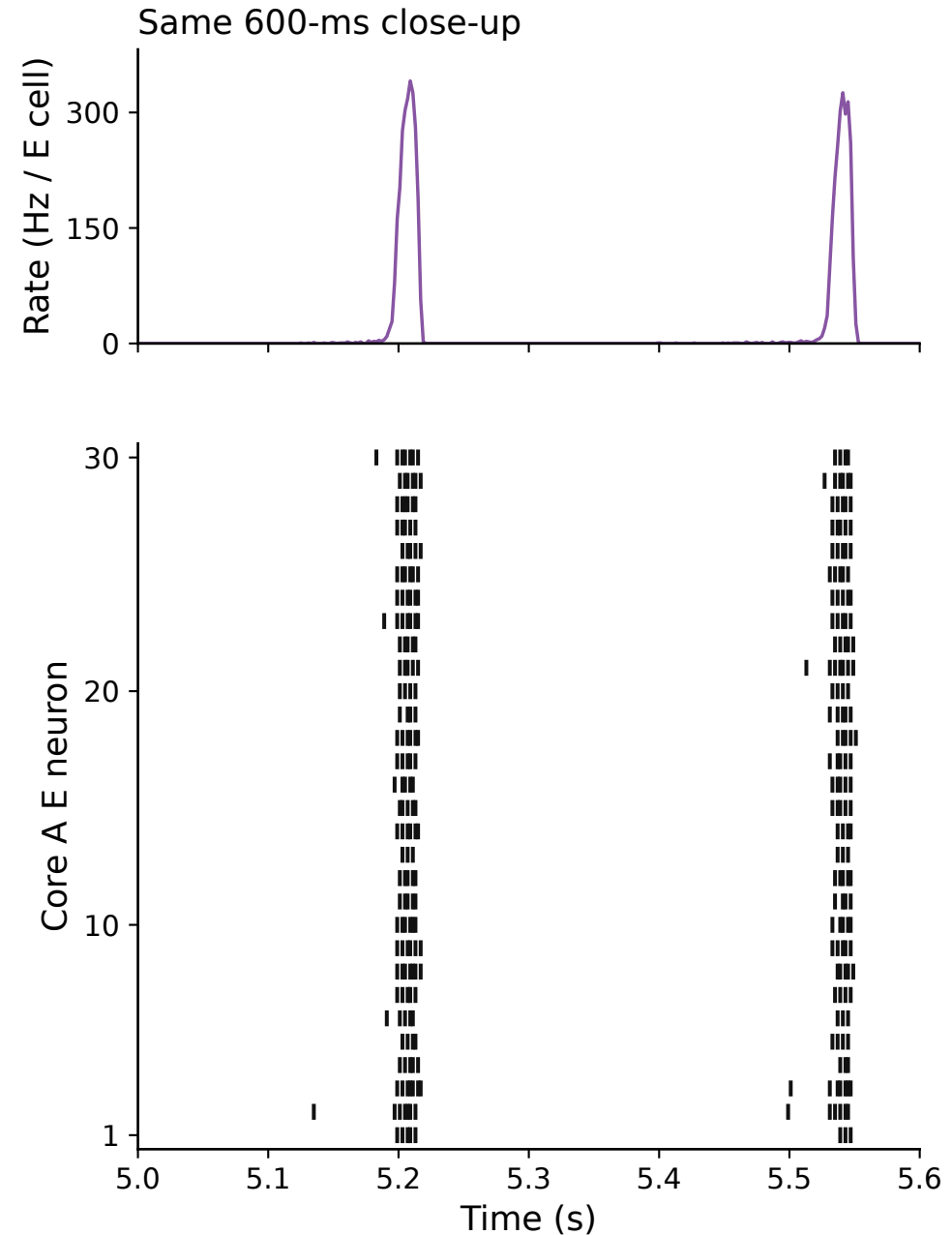
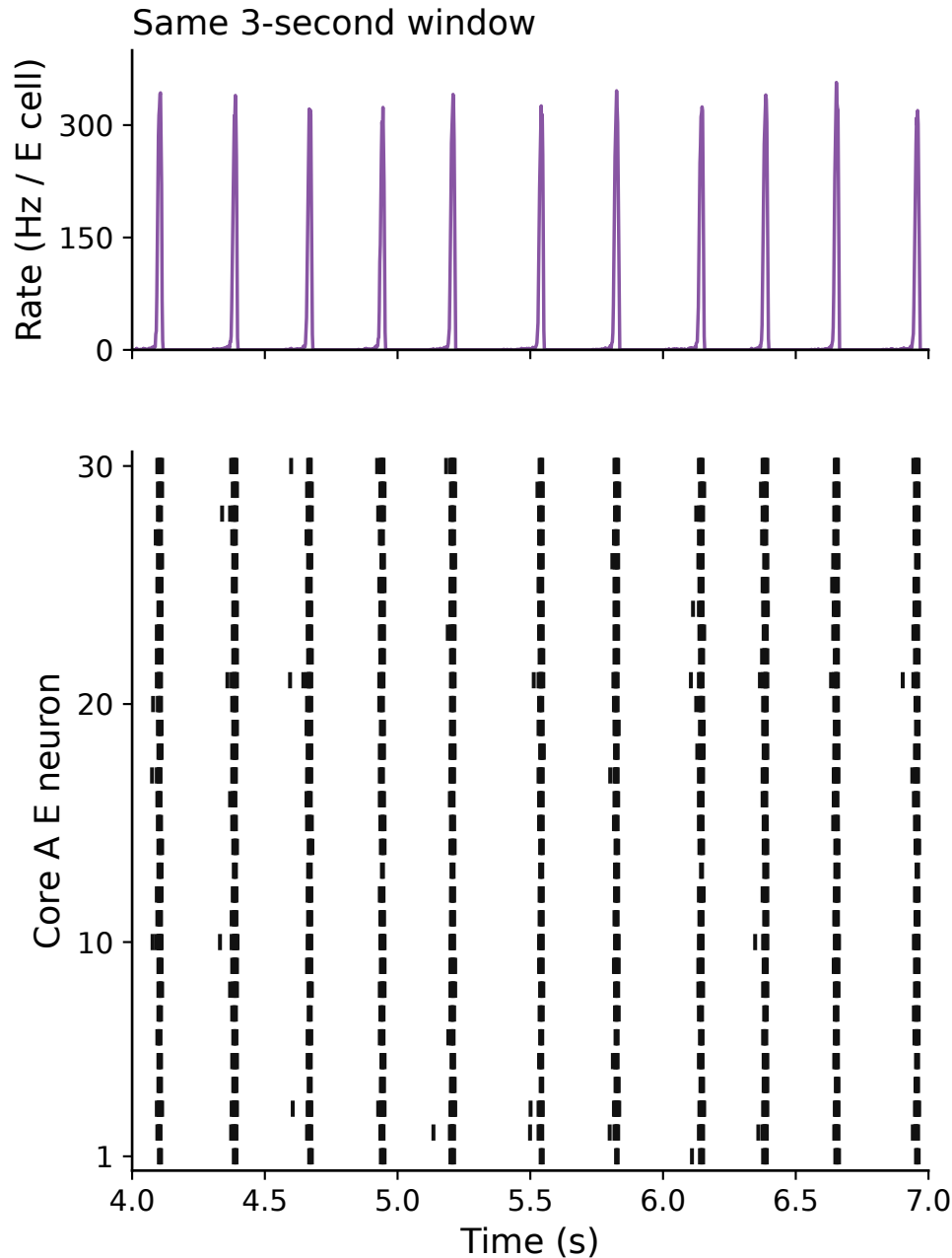


Native SNN. Rate: all 720 Core A E cells. Raster: occupied 2 ms bins of the same 30 recorded cells.

Metrics: 2-20 s; fixed topology 2511, threshold amplitude 1 and noise seed 848101.

4 Regular bursts

$J_{EE, core} = 1$ | IEI CV = 0.128; 62 bursts | mean = 18.887 Hz



Native SNN. Rate: all 720 Core A E cells. Raster: occupied 2 ms bins of the same 30 recorded cells.

Metrics: 2-20 s; fixed topology 2511, threshold amplitude 1 and noise seed 848101.